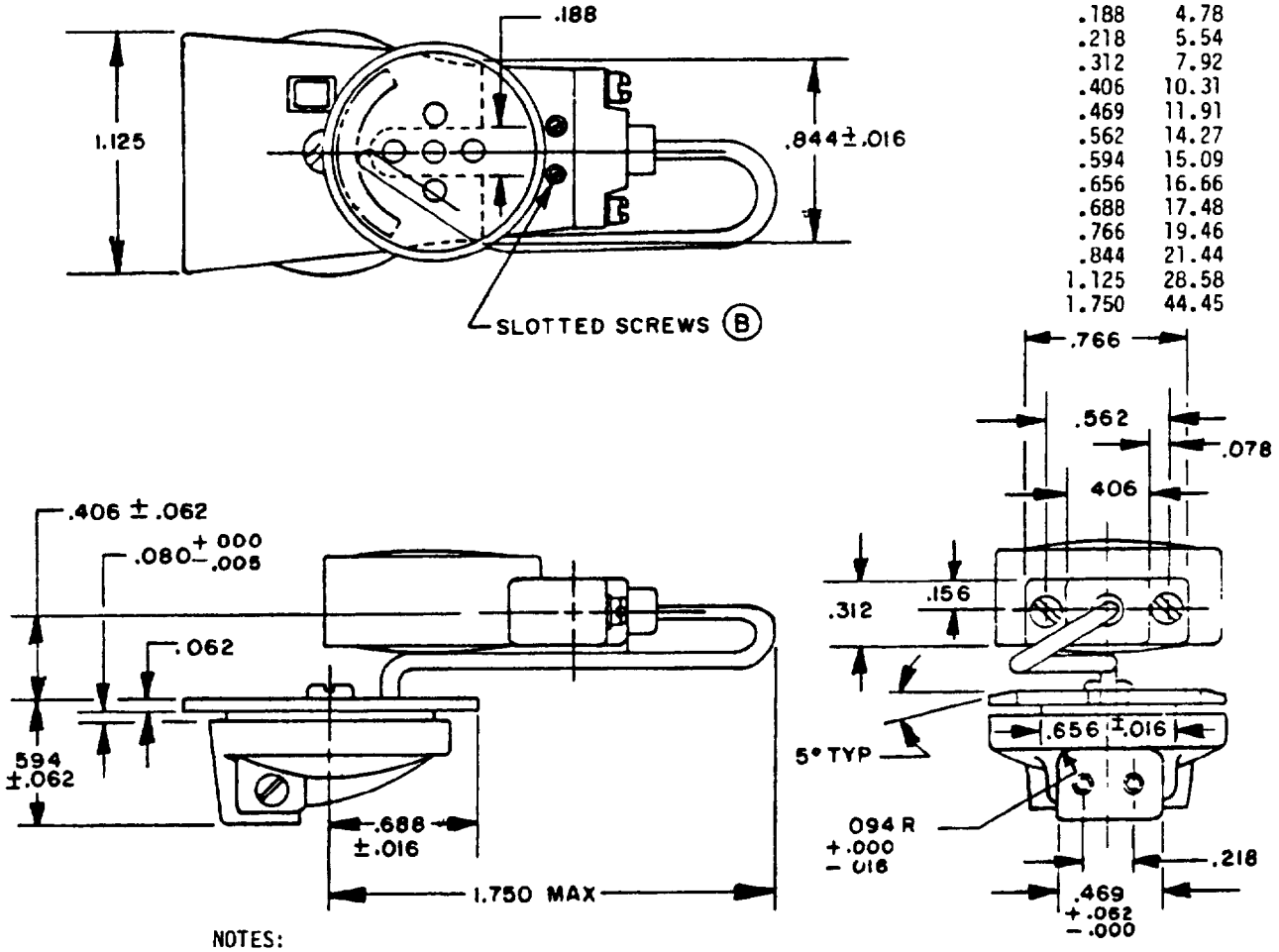
MILITARY SPECIFICATION SHEET

MICROPHONE, DYNAMIC M-102/AIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.

INCHES	MM
.005	0.13
.016	0.41
.062	1.57
.078	1.98
.080	2.03
.094	2.39
.156	3.96
.188	4.78
.218	5.54
.312	7.92
.406	10.31
.469	11.91
.562	14.27
.594	15.09
.656	16.66
.688	17.48
.766	19.46
.844	21.44
1.125	28.58
1.750	44.45



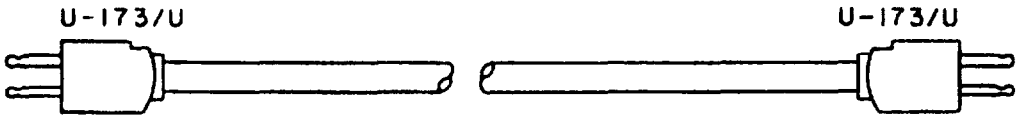
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

(B) FIGURE 1. Microphone assembly.

(B) denotes changes

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(B) FIGURE 2. Cable assembly M22442/36-5.

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REQUIREMENTS:

Positioning adjustment of bracket: The flexible tubing between the oxygen mask mounting plate and terminal block for the microphone element shall have a flex-life of at least 25 positioning adjustments without breaking or causing a short circuit between the internal conductors.

Mechanical fit to oxygen mask. The microphone bracket shall not permit any leakage between the face cavity and the microphone bracket cavity when installed in the MBU-4/P oxygen mask, or equal.

Functional operation: The microphone shall operate satisfactorily when subjected to the following functional operation test.

Functional operation test. Each microphone shall be given an operational test as a final test prior to packaging. The tester shall talk separately into the front and back sound ports of the microphone with equal loudness and with lips touching the moisture barrier guard. The tester shall wear good quality headset which is connected to the microphone through a suitable amplifier and shall listen to his own voice to insure that the microphone is functioning in a satisfactory manner. The amplifier used in this test shall not contain automatic gain control or other circuits that will compensate for low microphone output.

Color Semigloss black, 27038 in accordance with FED-STD-595.

Component parts. The microphone shall be supplied with the following components assembled:

Microphone element. One each M-101/AIC in accordance with MIL-M-26542/4.

Microphone bracket: One each MT-2190/AIC in accordance with figure 1.

(B) Cable assembly. M22442/36-5 (see figure 2).

Weight: The weight of microphone M-102/AIC shall not exceed 40.0 grams.

(B) Information sheet: An information sheet for microphone M-102/AIC shall be furnished with each microphone when it is packaged for delivery. The information sheet shall be approved by the Government activity responsible for qualification testing prior to quantity printing. The information sheet shall include information indicating that the "M-102/AIC" nomenclature is not required to be marked anywhere on the assembly but that the end item shall consist of the M-101/AIC microphone element and the appropriate mounting bracket which are individually marked with their own nomenclature for repair purposes.

Intended use: Microphone M-102/AIC is a noise-canceling dynamic moving-coil microphone designed for use with an oxygen mask, type MBU-4/P. The microphone is intended for use with intercommunication sets AN/AIC-10( ) and AN/AIC-18 to provide communication under the noise conditions encountered in military aircraft.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES

User activities:
Army - AR, AT, AV, ME
Navy - OS

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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